Principles of Anatomy and Physiology

Gerard Tortora and Bryan Derrickson

Sixteenth Edition

Chapter 5

The Integumentary System

Introduction

- The purpose of the chapter is to:

- Describe the structure and functions of the skin
  - Layers of epidermis and dermis
  - Types of cells in skin

- Distinguish accessory structures and different gland of skin

- Describe the anatomy of hair and nails

Animation

[The Integumentary System]

Maintenance of Homeostasis

Focus on Homeostasis

Contributions of the Integumentary System for All Body Systems

- **Skin and hair provide barriers that protect all internal organs from damaging agents in external environment**
- **Sweat glands and skin blood vessels regulate body temperature, needed for proper functioning of other body systems**

Skeletal System

- Skin helps activate vitamin D, needed for proper absorption of dietary calcium and phosphorus to build and maintain bones

Muscular System

- Skin helps provide calcium ions, needed for muscle contraction

Nervous System

- Nerve endings in skin and subcutaneous tissue provide input to brain for touch, pressure, thermal, and pain sensations

Endocrine System

- Keratinocytes in skin help activate vitamin D to calcitriol, a hormone that aids absorption of dietary calcium and phosphorus

Cardiovascular System

- Local chemical changes in dermis cause widening and narrowing of skin blood vessels, which help adjust blood flow to skin

Lymphoid (Lymphatic) System and Immunity

- Skin is first line of defense in immunity, providing mechanical barriers and chemical secretions that discourage penetration and growth of microbes
- Dendritic cells in epidermis participate in immune responses by recognizing and processing foreign antigens
- Macrophages in dermis phagocytize microbes that penetrate skin surface

Respiratory System

- Hairs in nose filter dust particles from inhaled air
- Stimulation of pain nerve endings in skin may alter breathing rate

Digestive System

- Skin helps activate vitamin D to the hormone calcitriol, which promotes absorption of dietary calcium and phosphorus in small intestine

Urinary System

- Kidney cells receive partially activated vitamin D hormone from skin and convert it to calcitriol
- Some waste products are excreted from body in sweat, contributing to excretion by urinary system

Genital (Reproductive) Systems

- Nerve endings in skin and subcutaneous tissue respond to erotic stimuli, thereby contributing to sexual pleasure
- Suckling of a baby stimulates nerve endings in skin, leading to milk ejection
- Mammary glands (modified sweat glands) produce milk
- Skin stretches during pregnancy as fetus enlarges

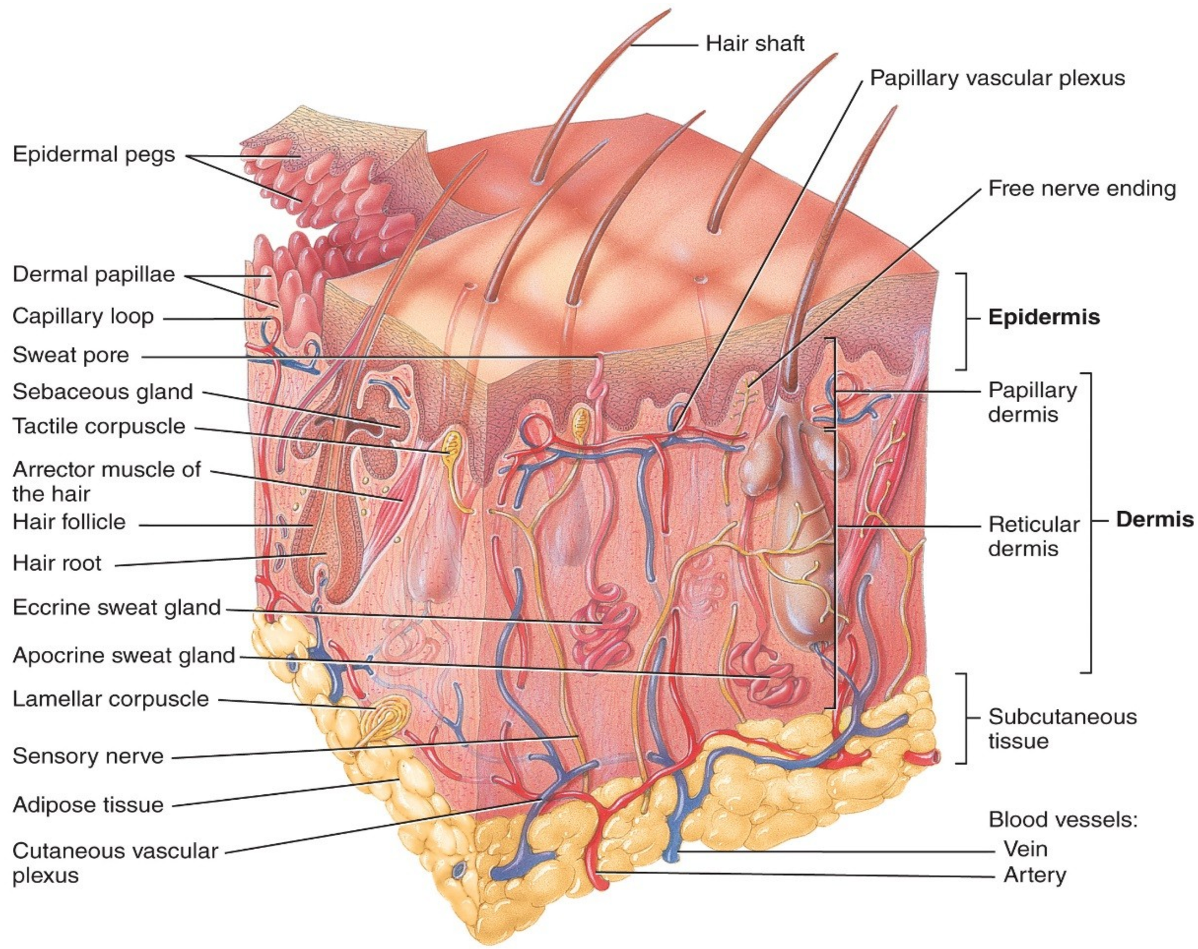
Structure of the Skin

Layers of the Skin

- The skin has 2 major layers:
  - Epidermis – the most superficial layer
  - Dermis – layer deep to the epidermis

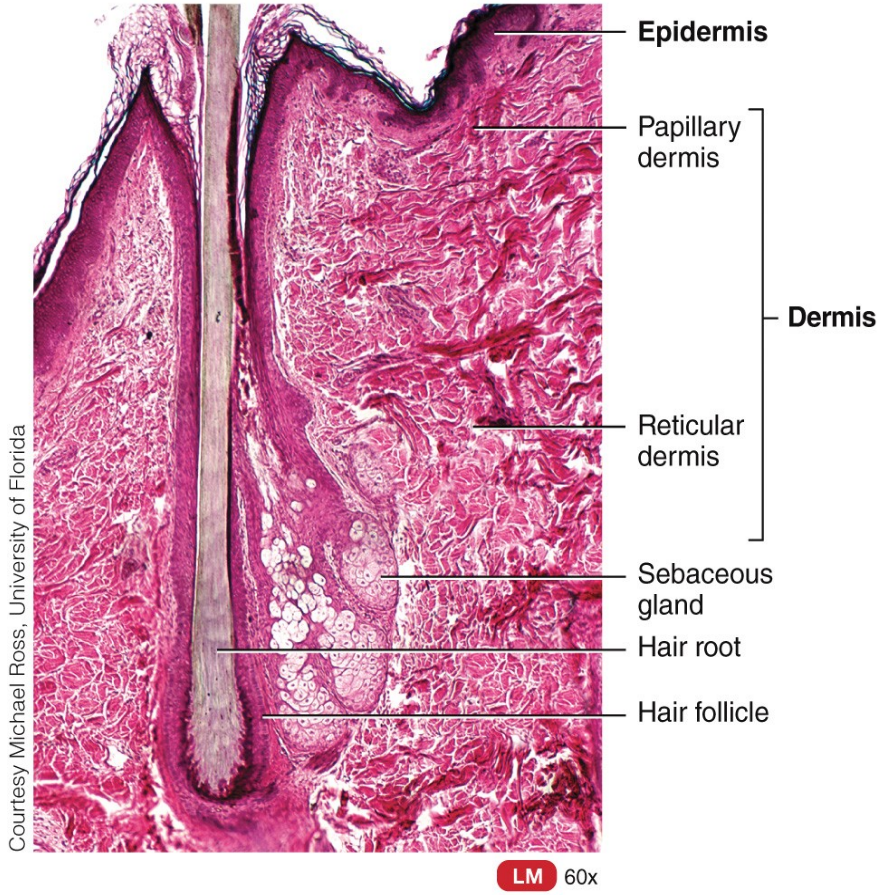
- Hypodermis – also called the subcutaneous (subQ) layer; located deep to the dermis but not a layer of the skin
  - Composed of areolar and adipose tissue

Components of the Integumentary System

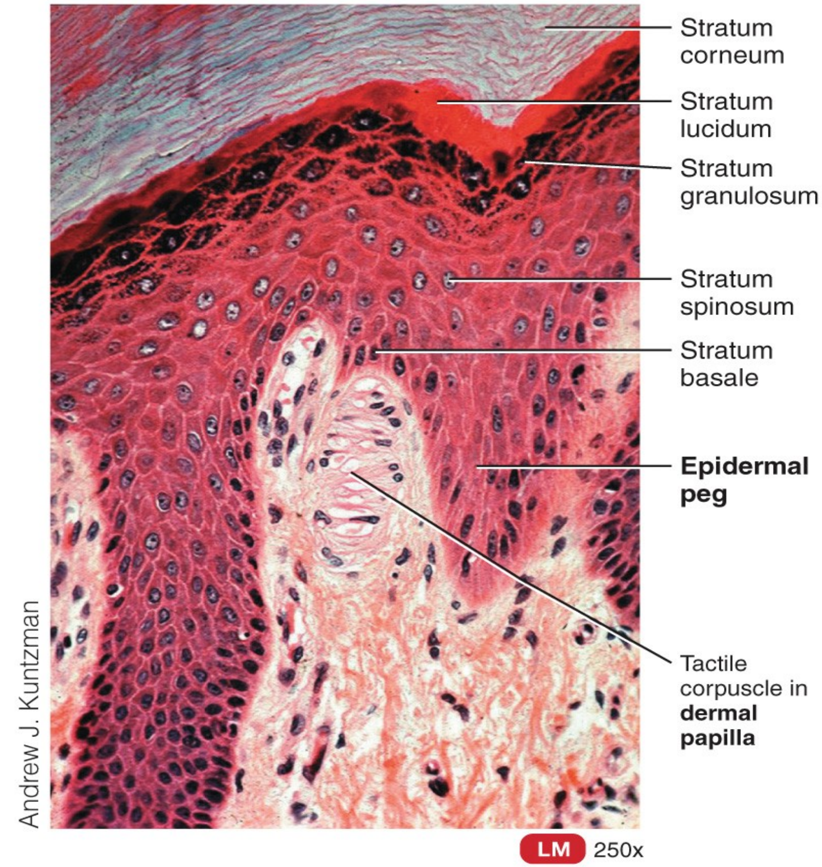


(a) Sectional view of skin and subcutaneous tissue

Components of the Integumentary System

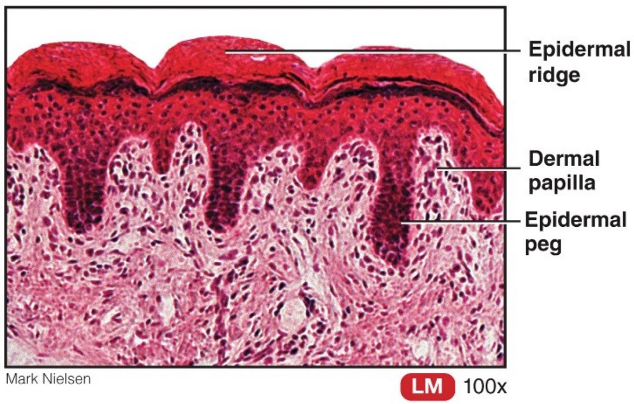


(b) Sectional view of skin

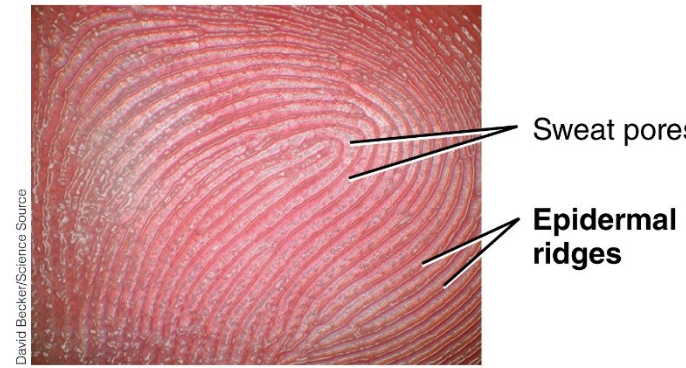


(c) Sectional view of dermal papillae, epidermal ridges, and epidermal layers

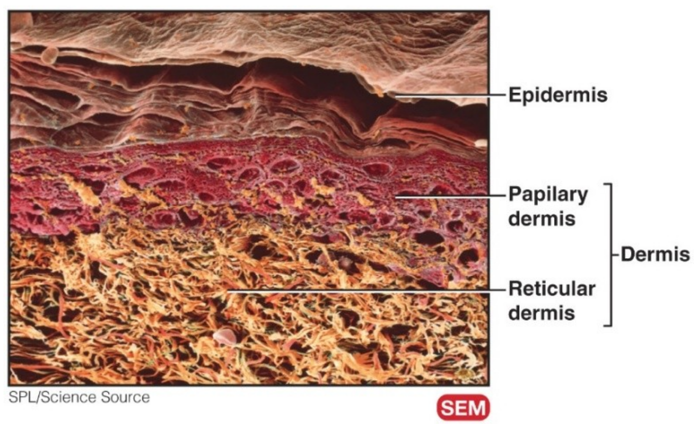
Components of the Integumentary System



(d) Relationship among epidermal ridge, dermal papilla, and epidermal peg



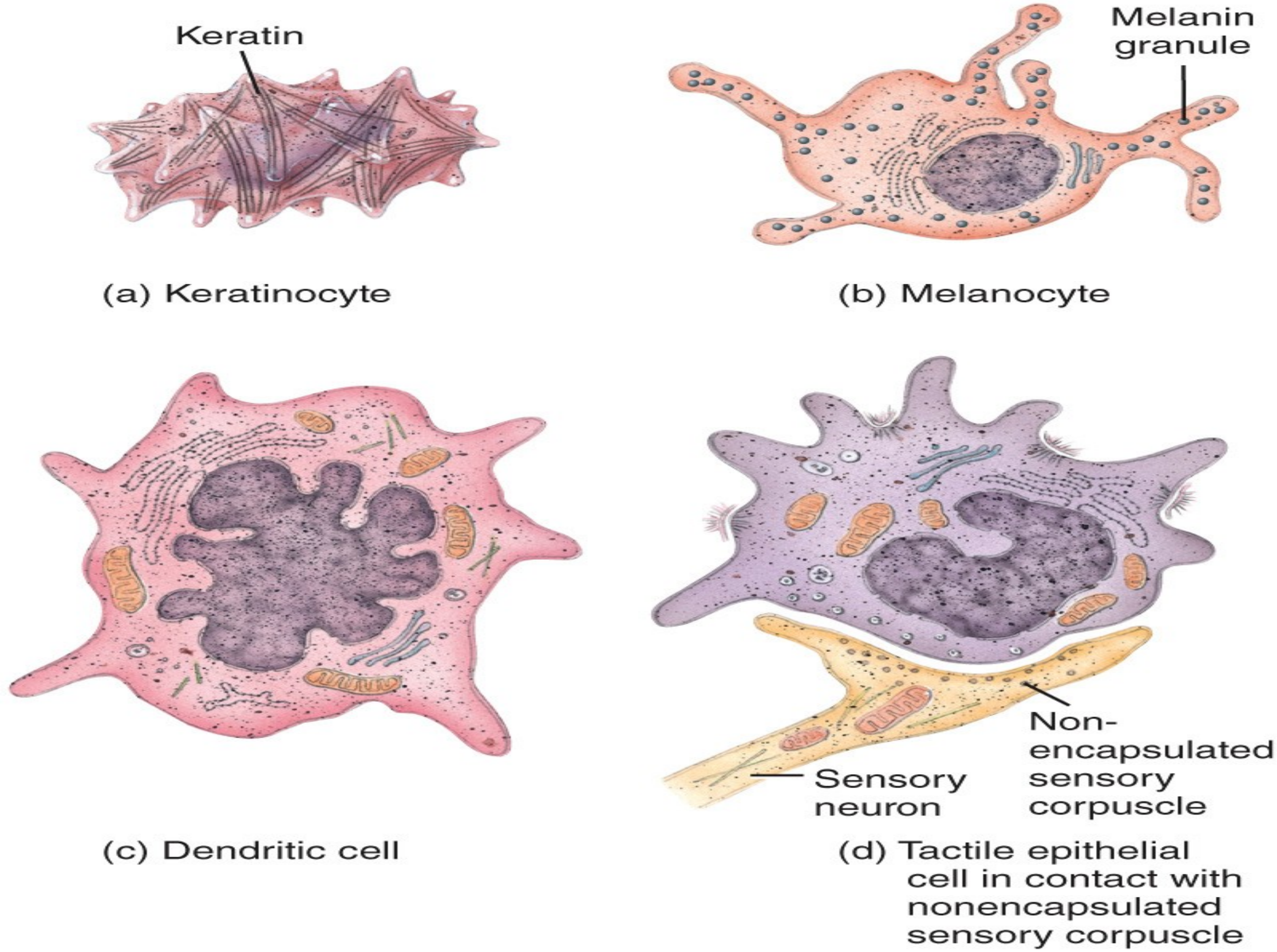
(e) Epidermal ridges and sweat pores



The Epidermis

- Four major types of cells
  - Keratinocytes
    - Produce fibrous protein keratin
  - Melanocytes
    - Produce melanin
  - Intradermal macrophages (Langerhans cells)
    - Helps activate immune system
  - Tactile epithelial cells (Merkel cells)
    - Touch receptors

The Epidermis



The Epidermis

- Types of skin:
  - Thin skin
    - Has hair
    - Covers all body regions except the palms, palmar surfaces of digits, and soles
  - Thick skin
    - Does not have hair
    - Covers palms, palmar surfaces of digits, and soles

TABLE 5.4 Comparison of Thin and Thick Skin

FEATURE	THIN SKIN	THICK SKIN
Distribution	All parts of body except areas such as palms, palmar surface of digits, and soles.	Areas such as palms, palmar surface of digits, and soles.
Epidermal thickness	0.10–0.15 mm (0.004–0.006 in.).	0.6–4.5 mm (0.024–0.18 in.), due mostly to a thicker stratum corneum.
Epidermal strata	Stratum lucidum essentially lacking; thinner strata spinosum and corneum.	Stratum lucidum present; thicker strata spinosum and corneum.
Epidermal ridges	Lacking due to poorly developed, fewer, and less-well-organized dermal papillae.	Present due to well-developed and more numerous dermal papillae organized in parallel rows.
Hair follicles and arrector pili muscles	Present.	Absent.
Sebaceous glands	Present.	Absent.
Sudoriferous glands	Fewer.	More numerous.
Sensory receptors	Sparser.	Denser.

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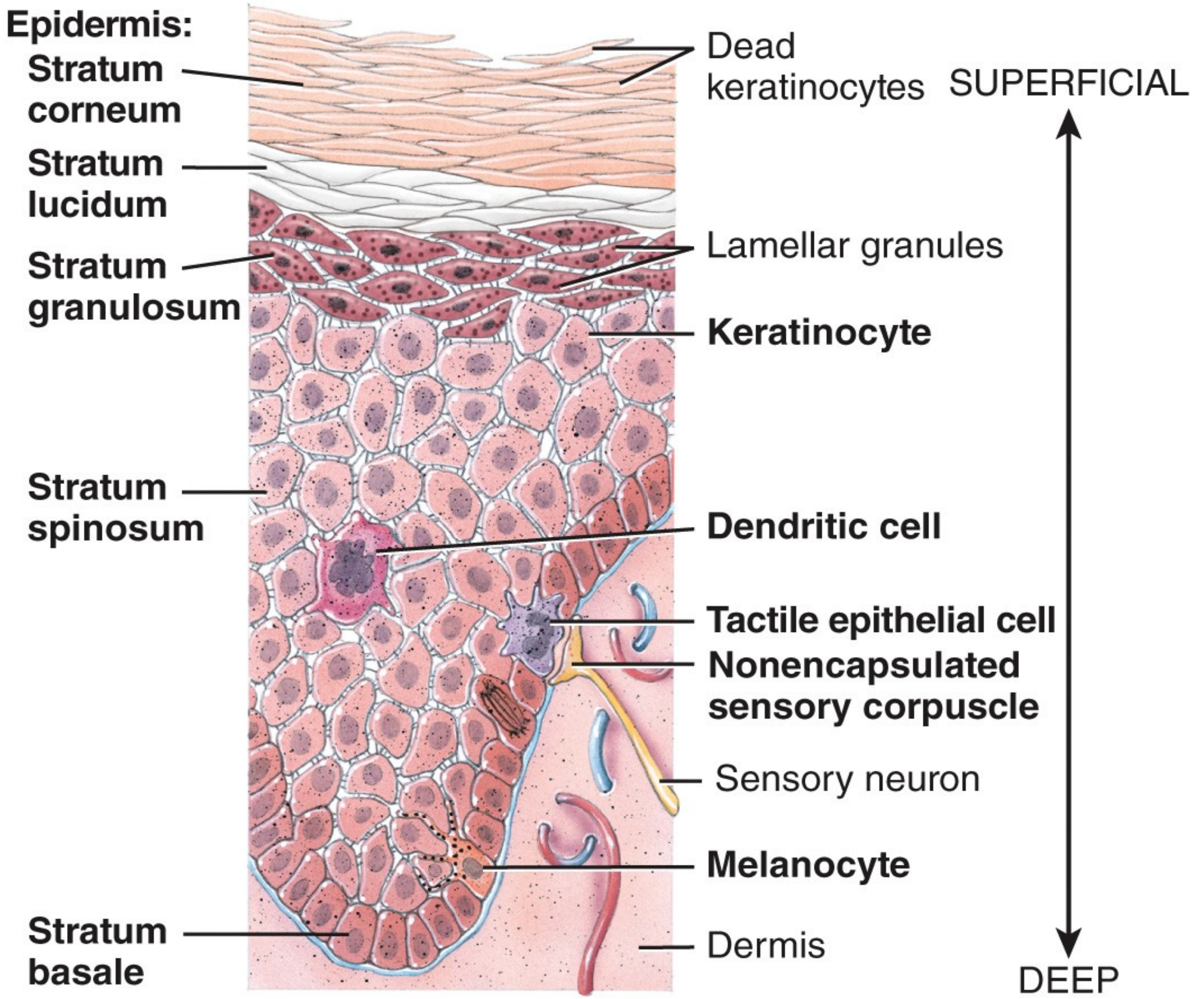
The Epidermis

- The epidermis is composed of four layers in thin skin and five layers in thick skin. They are (from deep to superficial):
  - The **stratum basale**
  - The **stratum spinosum**
  - The **stratum granulosum**
  - The **stratum lucidum** (only present in thick skin)
  - The **stratum corneum**

The Epidermis



Four principal cell types in epidermis of thick skin

TABLE 5.1 **Summary of Epidermal Strata (see Figure 5.3)**

STRATUM	DESCRIPTION
Basale	Deepest layer, composed of single row of cuboidal or columnar keratinocytes that contain scattered keratin intermediate filaments (tonofilaments); stem cells undergo cell division to produce new keratinocytes; melanocytes and tactile epithelial cells associated with tactile discs are scattered among keratinocytes.
Spinosum	Eight to ten rows of many-sided keratinocytes with bundles of keratin intermediate filaments; contains projections of melanocytes and intraepidermal macrophages.
Granulosum	Three to five rows of flattened keratinocytes, in which organelles are beginning to degenerate; cells contain the protein keratohyalin (converts keratin intermediate filaments into keratin) and lamellar granules (release lipid-rich, water-repellent secretion).
Lucidum	Present only in skin of fingertips, palms, and soles; consists of four to six rows of clear, flat, dead keratinocytes with large amounts of keratin.
Corneum	Few to 50 or more rows of dead, flat keratinocytes that contain mostly keratin.

Dermis

- Two layers
  - Papillary
    - Superficial; areolar connective tissue
    - Dermal papillae contain:
      - Capillary loops, Meissner's corpuscles, free nerve endings
  - Reticular
    - Deep; dense irregular connective tissue
    - Pacinian corpuscles
- Abundant blood supply – nutrients diffuse to deep layers of epidermis; waste diffused from epidermis

TABLE 5.2 Summary of Papillary and Reticular Regions of the Dermis (see Figure 5.1b)

REGION	DESCRIPTION
Papillary	Superficial portion of dermis (about one-fifth); consists of areolar connective tissue with thin collagen and fine elastic fibers; contains dermal ridges that house blood capillaries, corpuscles of touch, and free nerve endings.
Reticular	Deeper portion of dermis (about four-fifths); consists of dense irregular connective tissue with bundles of thick collagen and some coarse elastic fibers. Spaces between fibers contain some adipose cells, hair follicles, nerves, sebaceous glands, and sudoriferous glands.

Skin Pigments

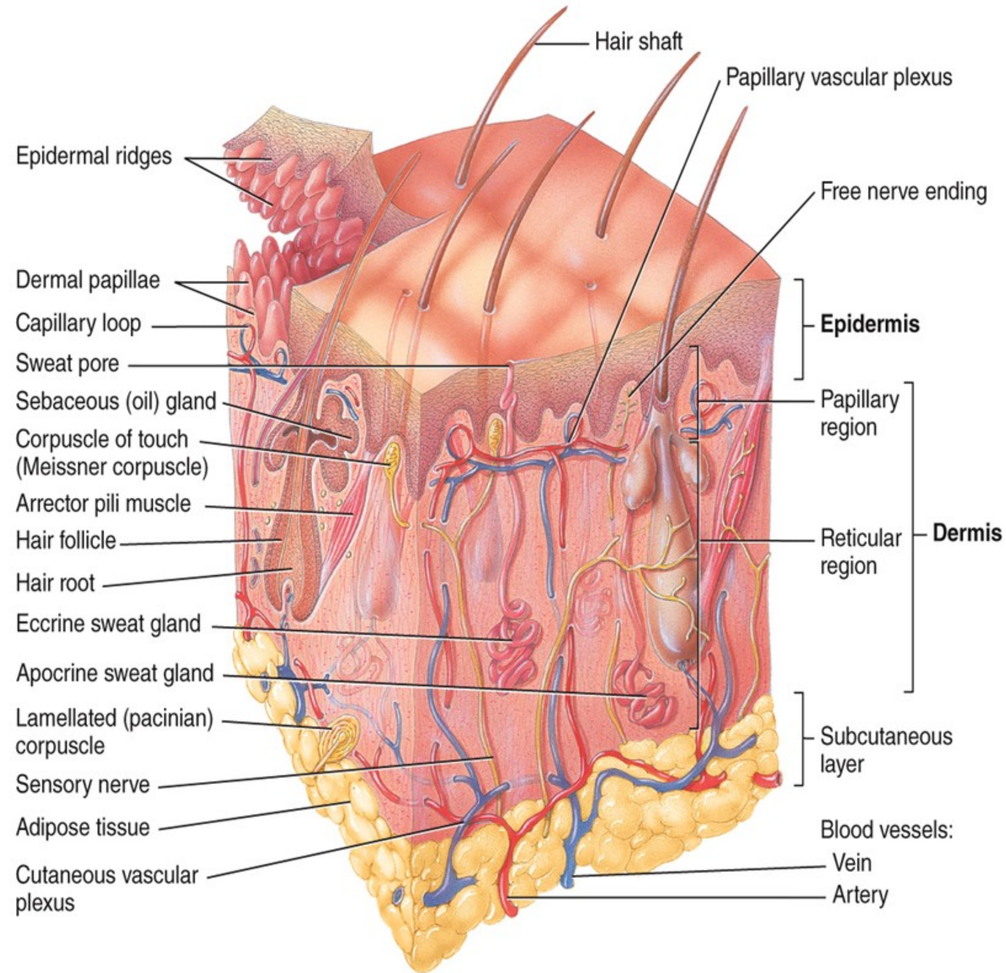
- Melanin - produced by melanocytes in the stratum basale
  - Pheomelanin (yellow to red)
  - Eumelanin (brown to black)
- Hemoglobin – red pigment in red blood cells
- Carotene – yellow-orange pigment stored in the stratum corneum and adipose tissue

Clinical Connection: Albinism and Vitiligo

- Albinism
  - Genetic trait characterized by complete or partial absence of pigment in skin, hair, and eyes
    - Due to enzyme involved with melanin production
- Vitiligo
  - Results in depigmentation patches in the skin
    - Exact cause is unknown; most likely due to combination of genetic factors coupled with a disfunction of the immune system

Hypodermis

- The subcutaneous layer
- Attaches skin to underlying tissues and organs



(a) Sectional view of skin and subcutaneous layer

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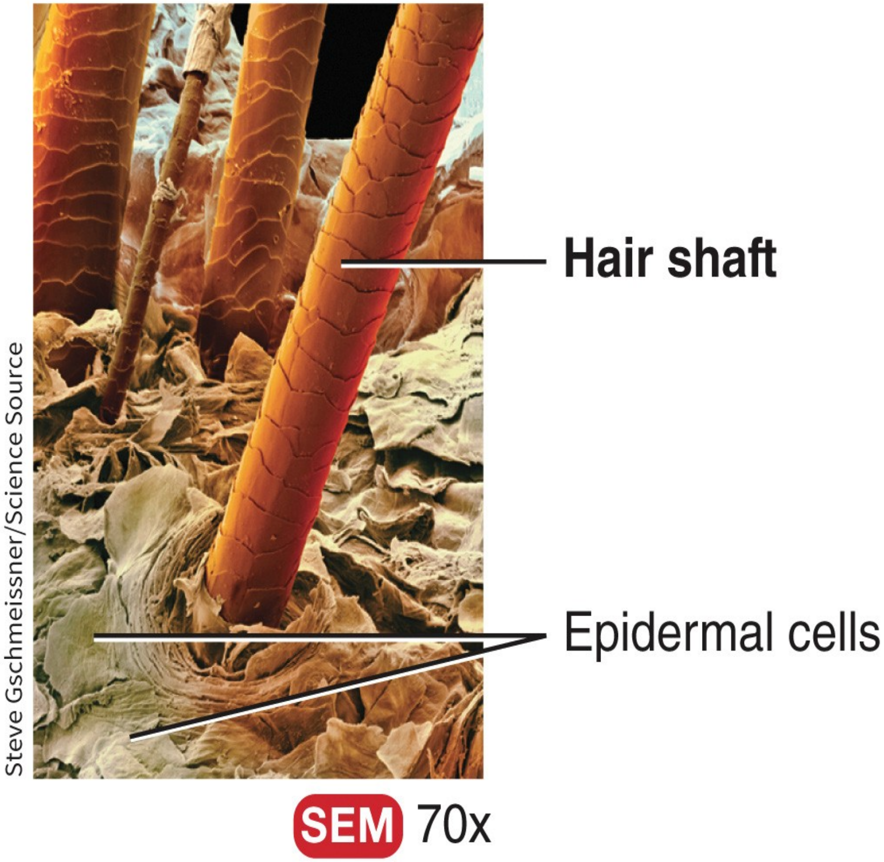
Accessory Structures of the Skin

Hair

- Present on most body surfaces except the nipples, the palms, palmar surfaces of fingers, the soles, plantar surfaces of the toes, labia minora and prepuce of penis
- Composed of dead, keratinized epidermal cells bonded by extracellular proteins
- Genetic and hormonal influences determine the thickness and distribution of our hair

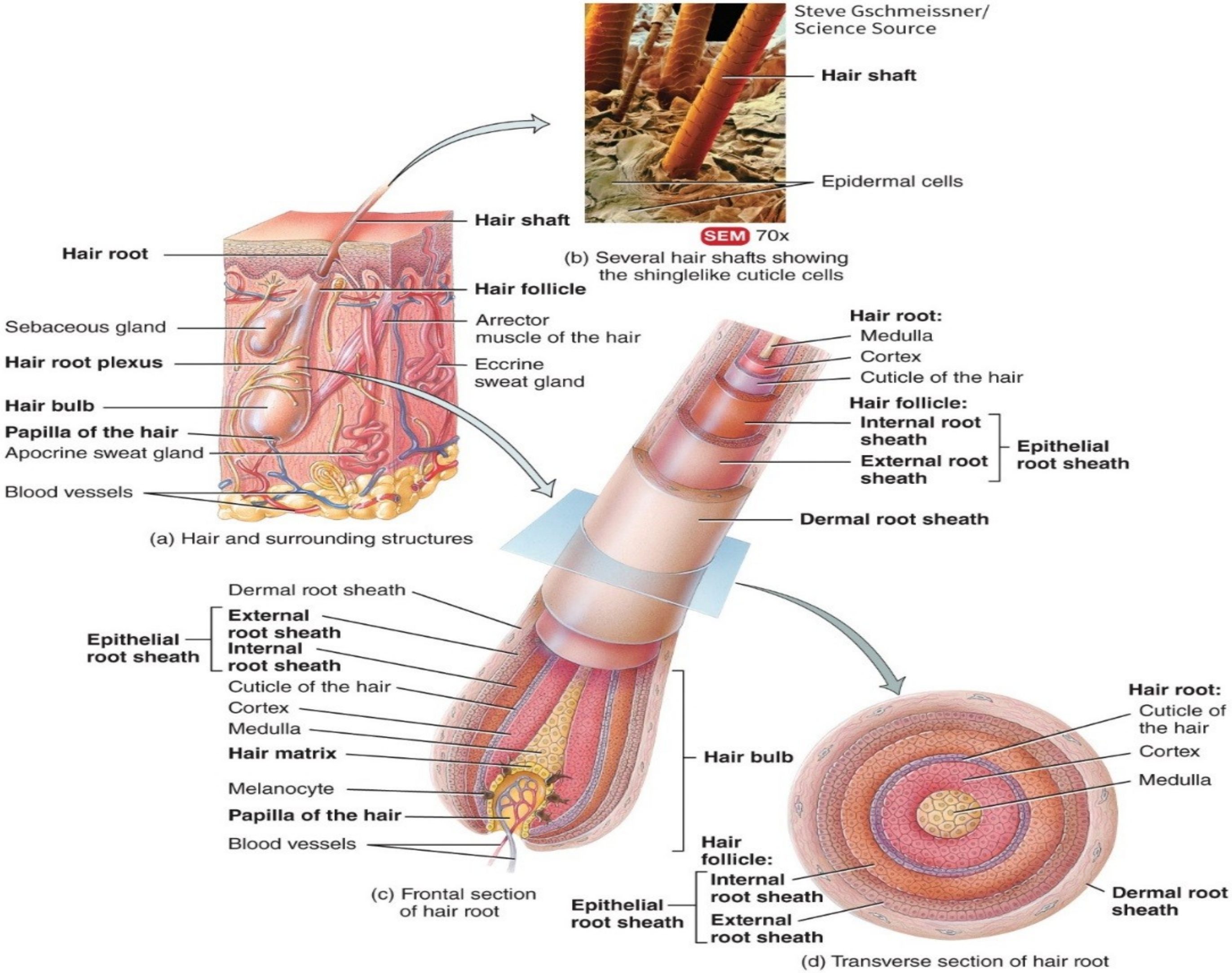
Hair

- The parts of a hair include:
  - The shaft (above the skin surface)
  - The follicle (below the level of the skin)
  - A root that penetrates into the dermis
    - Epithelial root sheath
    - Dermal root sheath



(b) Several hair shafts showing the shinglelike cuticle cells

Hair



Steve Gschmeissner/
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Hair shaft
Epidermal cells
SEM 70x
(b) Several hair shafts showing the shinglelike cuticle cells
Hair shaft
Hair root
Hair follicle
Sebaceous gland
Arrector muscle of the hair
Hair root plexus
Eccrine sweat gland
Hair bulb
Papilla of the hair
Apocrine sweat gland
Blood vessels
(a) Hair and surrounding structures
Hair root:
Medulla
Cortex
Cuticle of the hair
Hair follicle:
Internal root sheath
External root sheath
Epithelial root sheath
Dermal root sheath
Dermal root sheath
Epithelial root sheath
External root sheath
Internal root sheath
Cuticle of the hair
Cortex
Medulla
Hair matrix
Melanocyte
Papilla of the hair
Blood vessels
Hair bulb
(c) Frontal section of hair root
Hair root:
Cuticle of the hair
Cortex
Medulla
Hair follicle:
Internal root sheath
Epithelial root sheath
External root sheath
Dermal root sheath
(d) Transverse section of hair root

Hair Growth

- Hair growth stages are:
  - Growth stage – cells of matrix divide
  - Regression stage – hair moves away from blood supply in papillary and follicle atrophies
  - Resting stage – old hair root falls out; new growth begins

Types of Hairs

- Lanugo – cover the fetus
- Terminal – long, coarse, heavily pigmented hairs
- Vellus – short, fine, pale hairs

Hair Color

- Hair color is primarily due to the amount and type of melanin present in the keratinized cells of the hair
    - Dark-colored hair – mostly eumelanin
    - Blond and red hair – variants of phenomelanin
    - Gray hair – progressive decline in melanin production
    - White hair – lack of melanin; accumulation of air bubbles in shaft

Skin Glands

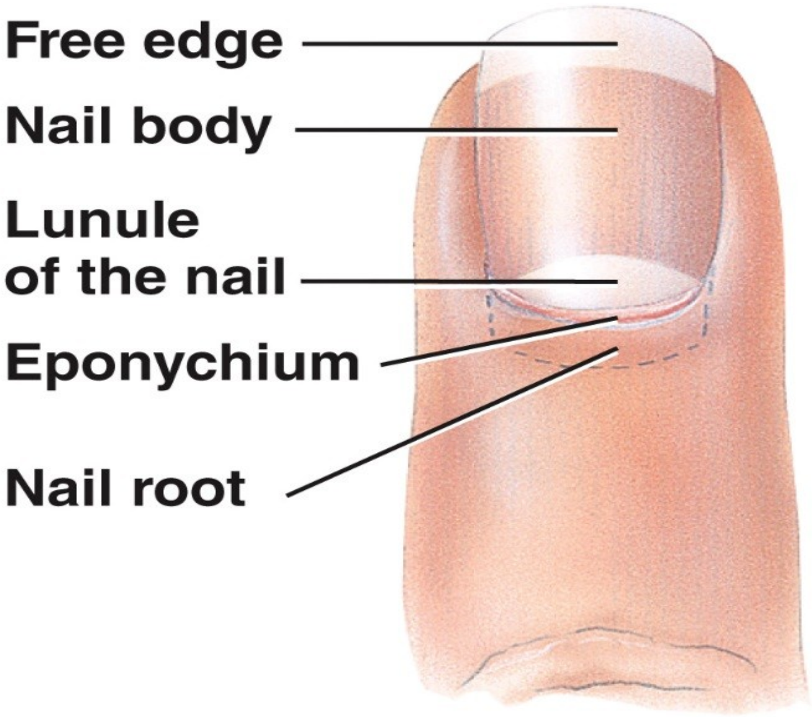
- The skin contains 3 types of glands
  - **Sebaceous (oil) glands** are connected to hair follicles
  - **Sudoriferous glands** are sweat glands
    - **Eccrine** are the most numerous
      - Ducts connect to pores
      - Function in thermoregulation
    - **Apocrine** are located mainly in hairy skin areas
      - Ducts connect to hair follicles
      - Axillary and pubic regions
  - **Ceruminous glands** are modified sweat glands located in the ear canal

TABLE 5.3 Summary of Skin Glands (see Figures 5.1 and 5.4a)

FEATURE	SEBACEOUS (OIL) GLANDS	ECCRINE SWEAT GLANDS	APOCRINE SWEAT GLANDS	CERUMINOUS GLANDS
Distribution	Largely in lips, glans penis, labia minora, and tarsal glands; small in trunk and limbs; absent in palms and soles.	Throughout skin of most regions of body, especially skin of forehead, palms, and soles.	Skin of axillae, groin, areolae, bearded regions of face, clitoris, and labia minora.	External auditory canal.
Location of secretory portion	Dermis.	Mostly in deep dermis (sometimes in upper subcutaneous layer).	Mostly in deep dermis and upper subcutaneous layer.	Subcutaneous layer.
Termination of excretory duct	Mostly connected to hair follicle.	Surface of epidermis.	Hair follicles.	Surface of external auditory canal or into ducts of sebaceous glands.
Secretion	Sebum (mixture of triglycerides, cholesterol, proteins, and inorganic salts).	Perspiration, which consists of water, ions (Na^+, Cl^-), urea, uric acid, ammonia, amino acids, glucose, and lactic acid.	Perspiration, which consists of same components as eccrine sweat glands plus lipids and proteins.	Cerumen, a waxy material.
Functions	Prevent hairs from drying out, prevent water loss from skin, keep skin soft, inhibit growth of some bacteria.	Regulation of body temperature, waste removal, stimulated during emotional stress.	Stimulated during emotional stress and sexual excitement.	Impede entrance of foreign bodies and insects into external ear canal, waterproof canal, prevent microbes from entering cells.
Onset of function	Relatively inactive during childhood; activated during puberty.	Soon after birth.	Puberty.	Soon after birth.

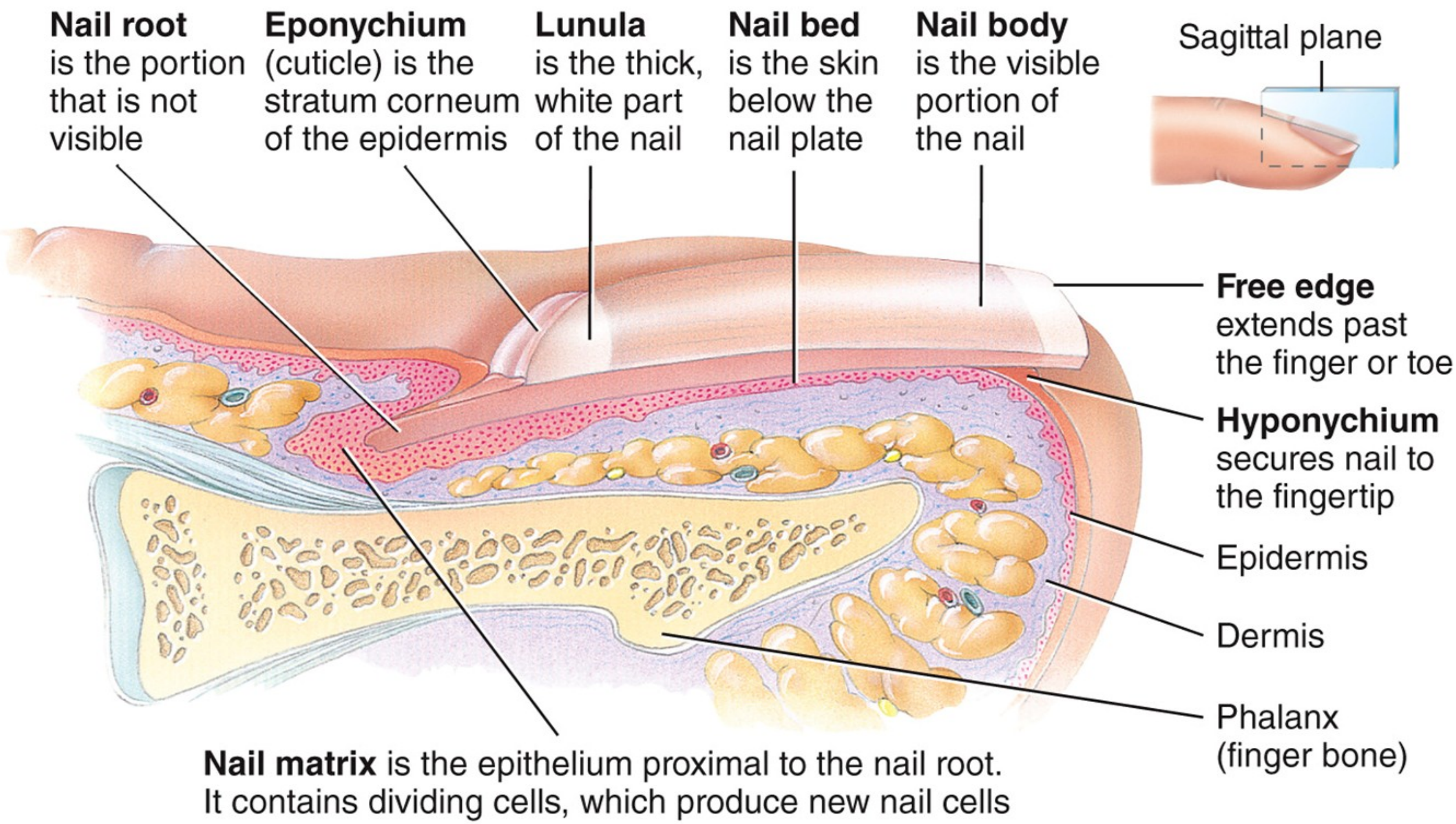
Nails

- Nails are made of keratinized epidermal cells



(a) Dorsal view

Nails



(b) Sagittal section showing internal detail

Functions of Skin

Functions of the Skin

- Thermoregulation
  - Sweat – evaporative cooling
  - Blood flow to dermis
- Blood reservoir
  - 8 – 10% of total blood flow due to abundant blood vessels in dermis

Functions of the Skin

- Protection
  - Keratin
  - Lipids released by lamellar granules
  - Sebum
  - Acidic sweat
  - Melanin
  - Macrophages
- Cutaneous sensations
  - Tactile sensations – touch, pressure, vibration, tickle
  - Thermal sensation – warm, cool
  - Pain

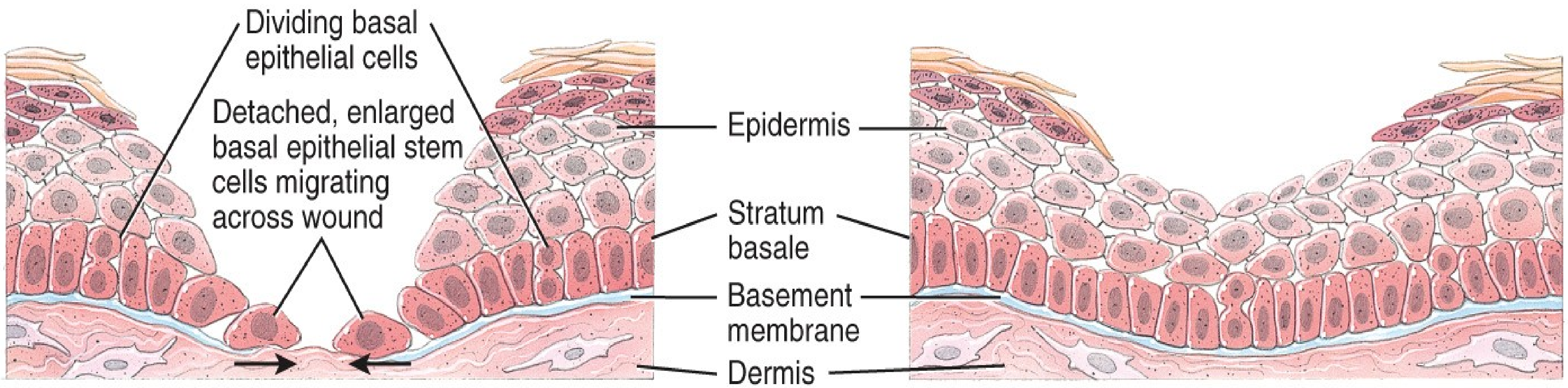
Functions of the Skin

- Excretion and absorption
  - Elimination of substances from the body and passage of materials from external environment into body cells
  - Transdermal drug administration
- Synthesis of Vitamin D
  - UV light activates precursor molecule (calcitriol) that allows vitamin D to be produced
    - Vitamin D aids in absorption of calcium in GI tract

Wound Healing

Epidermal Wound Healing

Occurs when superficial wounds affect only the epidermis



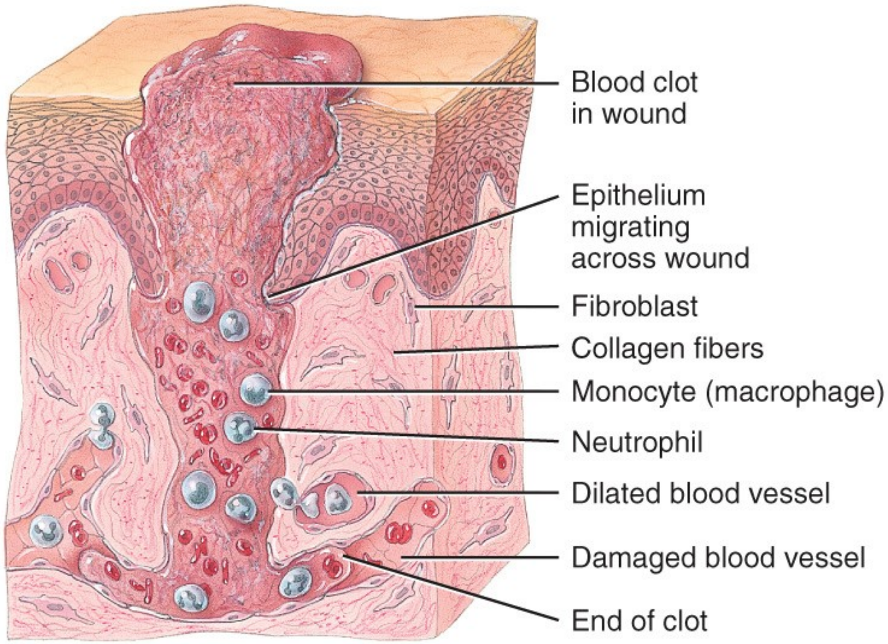
(a) Division of epidermal stem cells in the stratum basale and migration across wound

(b) Thickening of epidermis

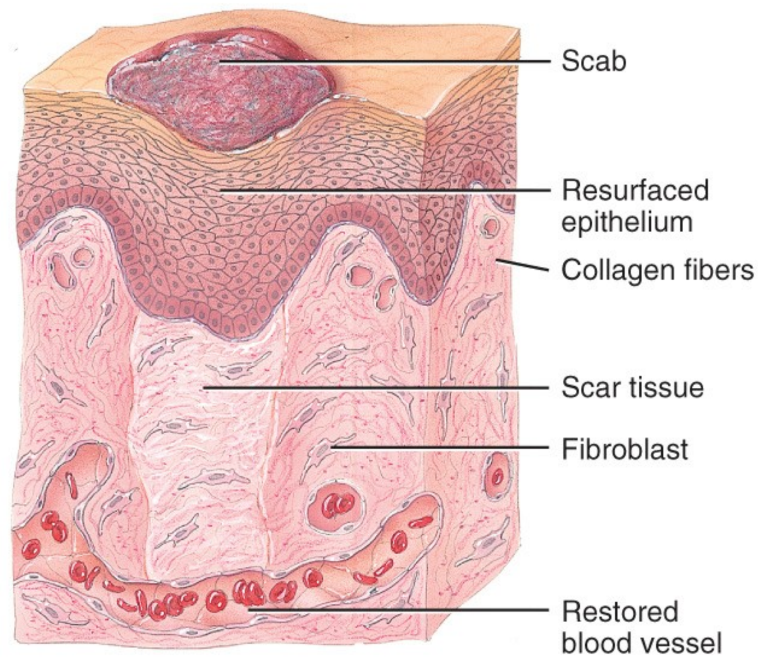
Epidermal wound healing

Deep Wound Healing

Additional steps occur when an injury extends into the dermis and subcutaneous layer



(c) Inflammatory phase



(d) Maturation phase

Deep wound healing

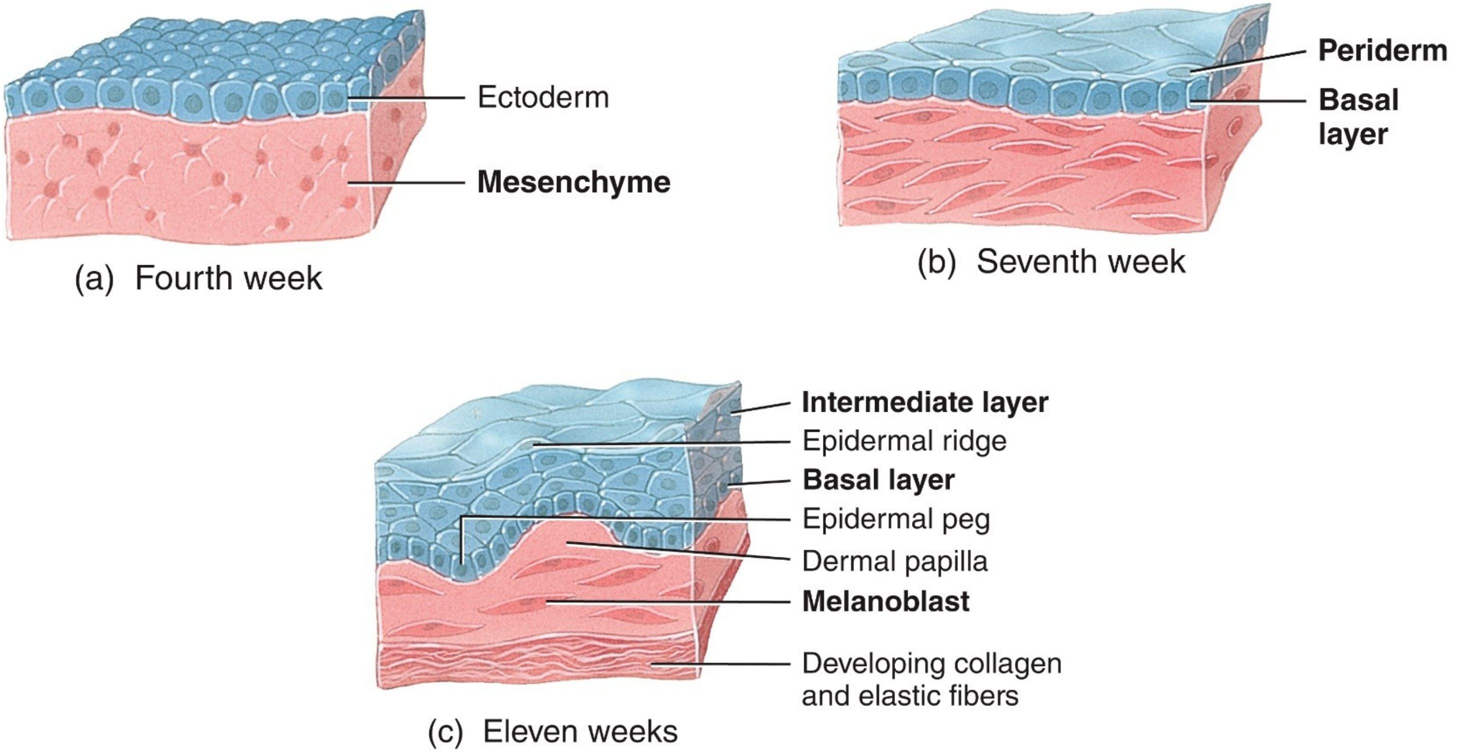
Deep Wound Healing

- Inflammatory Phase – clot forms
- Migratory Phase – clot becomes a scab
- Proliferative Phase – growth of epithelial cells beneath scab
- Maturation Phase – scab sloughs off once epidermis restored

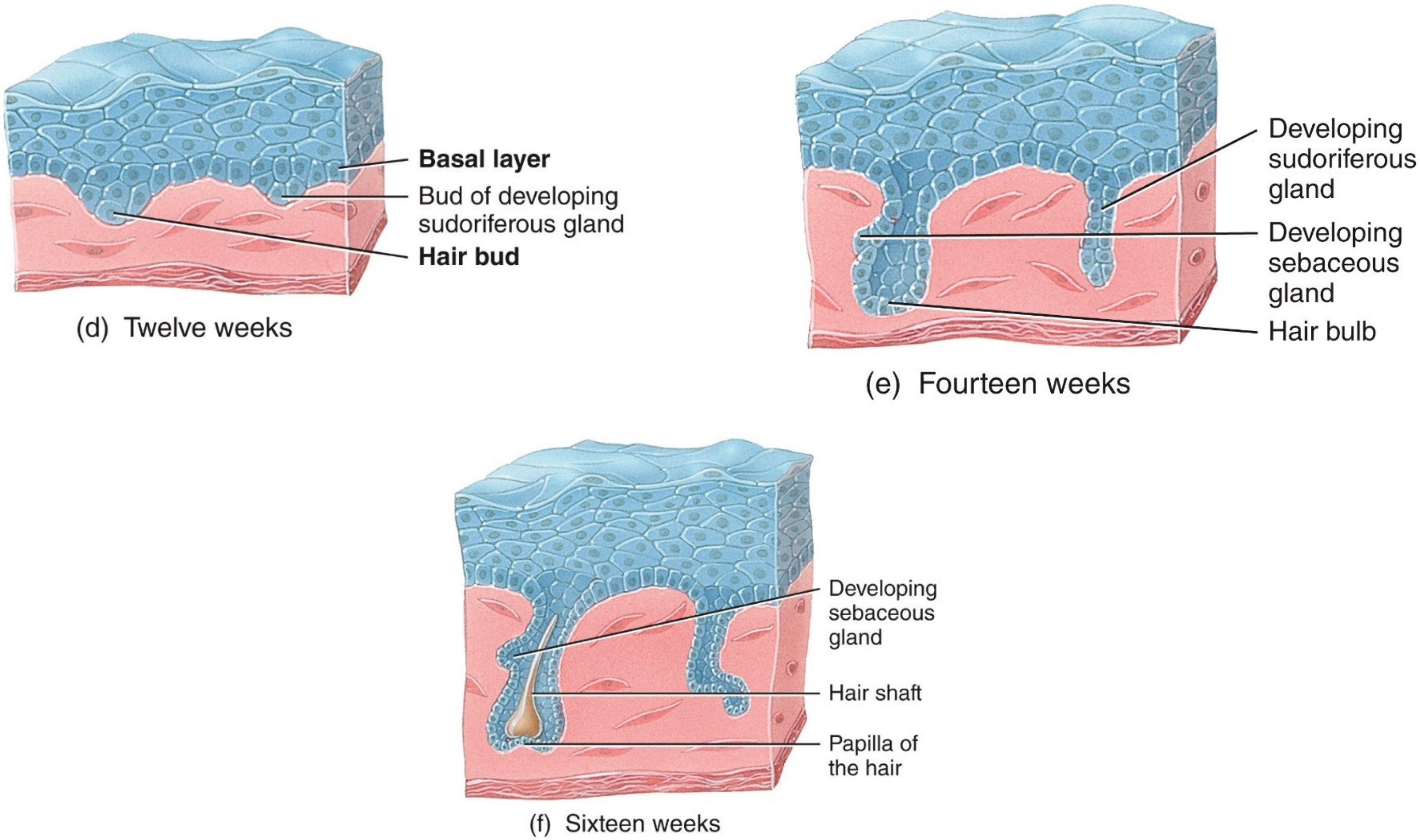
Development of the Integumentary System

Epidermis Develops from Ectoderm

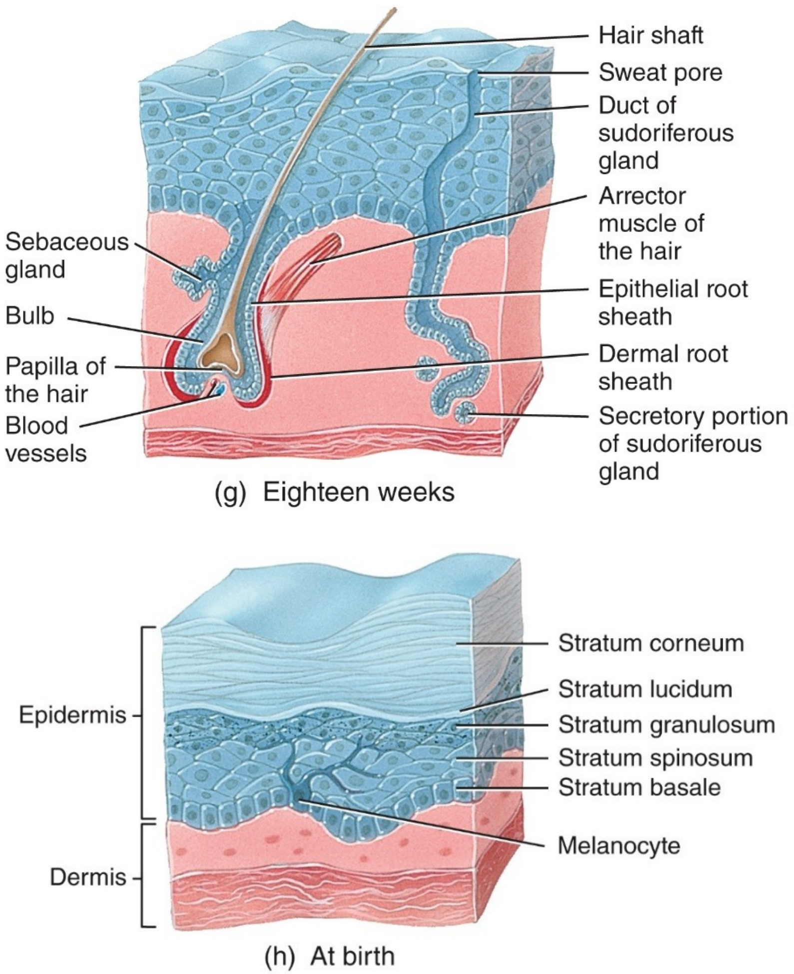
Nails, hair, and skin glands are epidermal derivatives



Epidermis Develops from Ectoderm

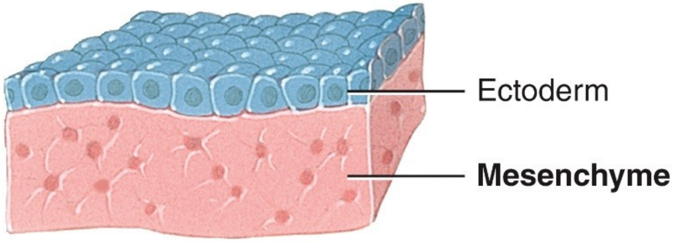


Epidermis Develops from Ectoderm

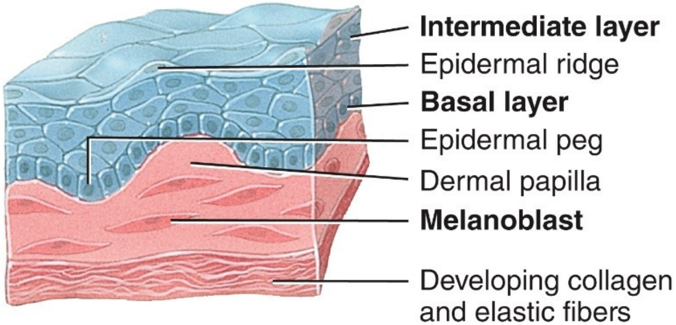


Dermis Develops from Mesoderm

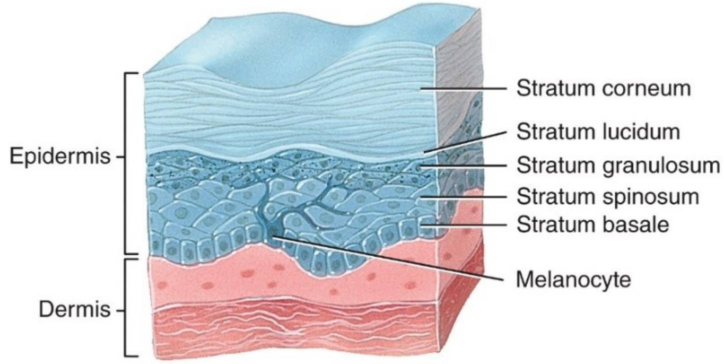
Mesoderm gives rise to mesenchyme



(a) Fourth week



(c) Eleven weeks



(h) At birth

Aging

Age-Associated Changes

- Wrinkles
- Dehydration and cracking
- Sweat production decreases
- The numbers of functional melanocytes decrease which results in gray hair and atypical skin pigmentation
- Subcutaneous fat is lost and skin thickness decreases
- Nails may become more brittle

Clinical Connection: Skin Cancer

- Excessive exposure to ultraviolet light (from the sun or tanning salons) is the most common cause of skin cancer

- The three major types are basal cell carcinoma (78%), squamous cell carcinoma (20%) and malignant melanoma (2%)

Clinical Connection: Skin Cancer

(a) Normal nevus (mole)

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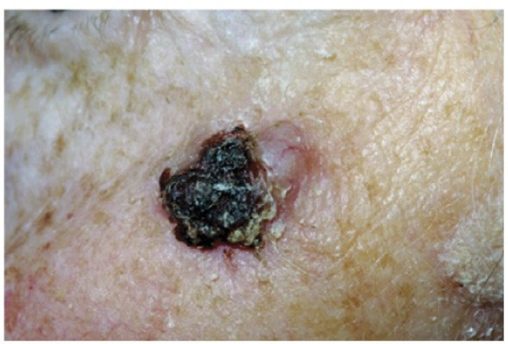

(b) Basal cell carcinoma

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(c) Squamous cell carcinoma

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(d) Malignant melanoma

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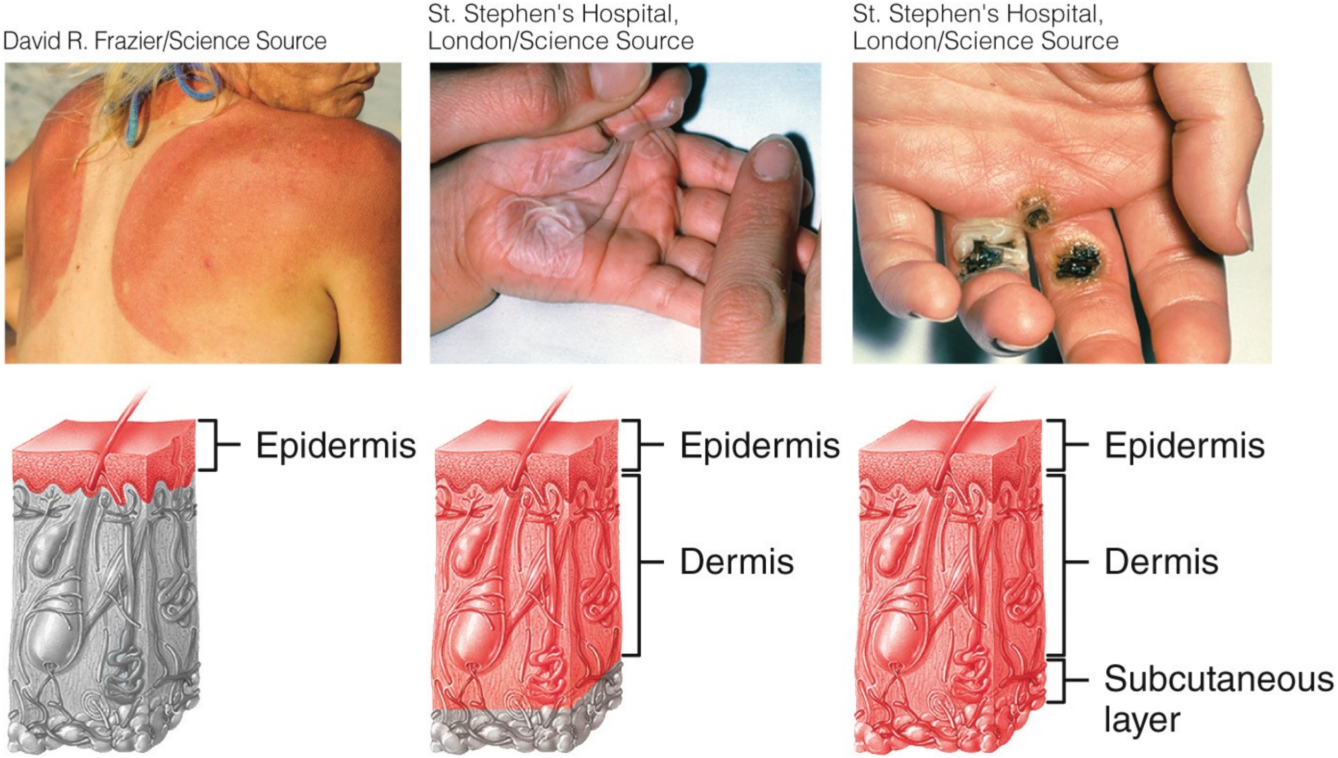
Clinical Connection: Skin Cancer

- Early warning signs of malignant melanoma
  - A – asymmetry
  - B – border is irregular
  - C – color is uneven
  - D – diameter is >6mm (pencil eraser)
  - E – evolving and changing in size and shape

Clinical Connection: Burns

- A **burn** is tissue damage caused by excessive heat, electricity, radioactivity, or corrosive chemicals that denature (break down) the proteins in the skin cells
- Burns are graded according to their severity

Clinical Connection: Burns



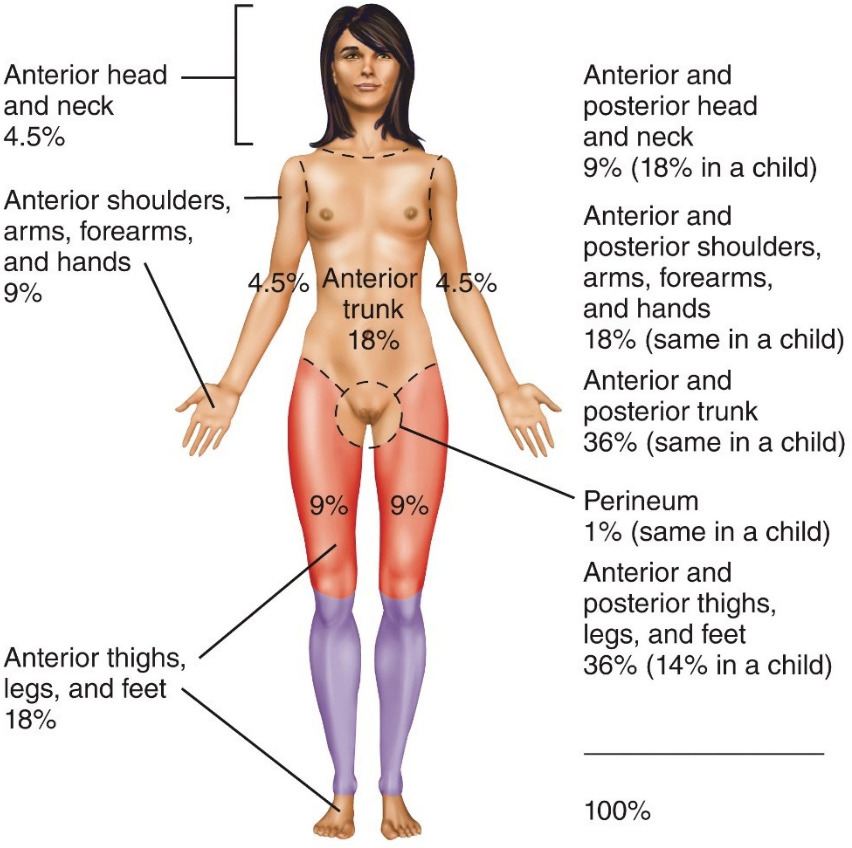
(a) First-degree burn (sunburn)

(b) Second-degree burn (note the blisters in the photograph)

(c) Third-degree burn

Clinical Connection: Burns

The rule of nines is used to estimate the surface area of an adult affected by a burn



(D) Anterior view illustrating rule of nines

Clinical Connection: Pressure Ulcers

- With age, there is an increased susceptibility to pressure ulcers ("bed sores")
- When shedding of epithelium caused by a deficiency of blood flow to tissues occurs, pressure ulcers can develop

Pressure ulcer on heel

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